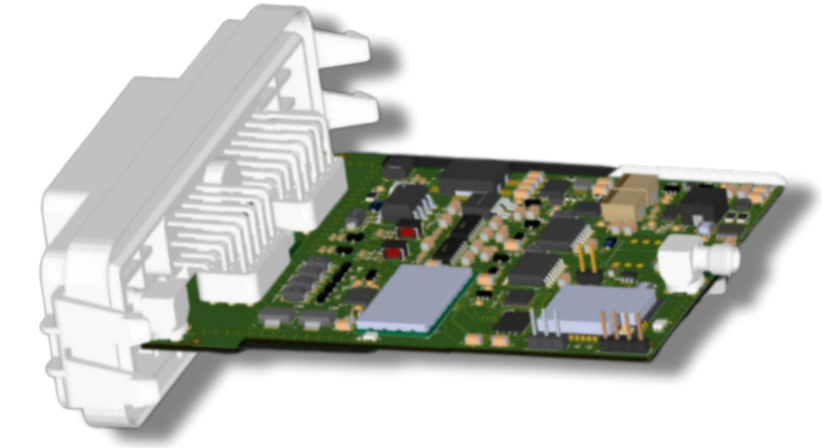
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IOT CAN GATEWAY - ENEDYM (CO-OP SUMMER 2026)





What?

- Independently designed the complete hardware for an in-house **IoT CAN Bus Data Logger**, **cutting per-device cost by over $500** while improving field reliability and connectivity.

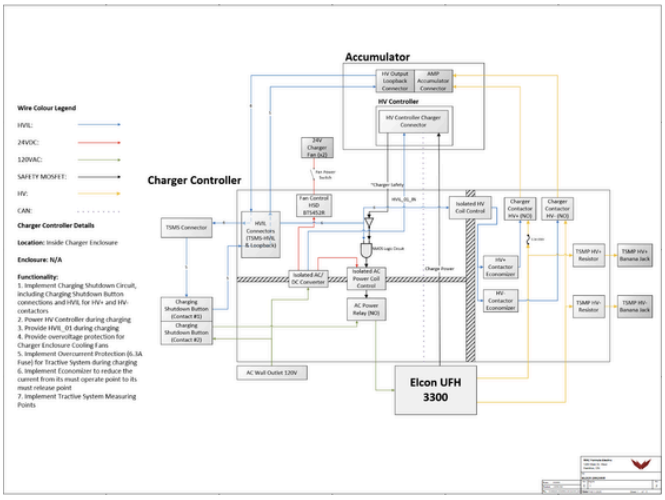
How?

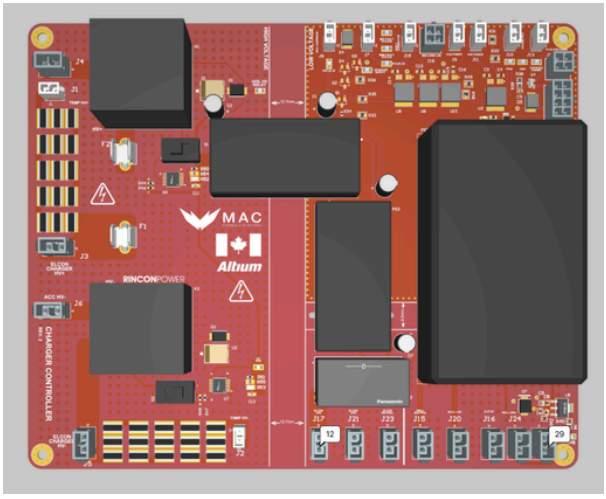
- Designed **6-layer PCB** in **Altium Designer**, integrating a **microcontroller**, **cellular/GNSS module**, and **CAN transceivers.**
- **Developed C firmware** for **real-time CAN bus acquisition**, **decoding**, and **wireless telemetry upload.**

Results

- **Delivered** a sealed, rugged **device that streams live CAN telemetry over cellular/Wi-Fi** and buffers data locally until reconnected.
- **Cut per-device cost by over $500** versus the off-the-shelf unit.

CHARGER REV3 PCB - MAC FORMULA ELECTRIC





What?

- Led charger controller redesign with **13 hardware improvements**, enhancing **safety**, **efficiency**, and **communication reliability**.

How?

- Simulated the **HVIL/charger-safety NMOS logic** and found a missed truth-table case
- **Reworked the REV2** board by hand: removed a FET, bridged its source–drain, and fly-wired a new PMOS
- In REV3, **moved** power to **onboard AC-DC**, **added an HV-fuse**, and fixed the **relay DC-DC wiring**

Results

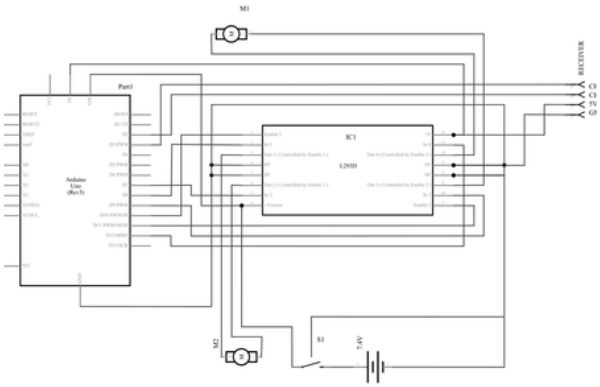
- **Corrected the charger-safety logic** in REV2 and carried the fixes into the **REV3 board,**

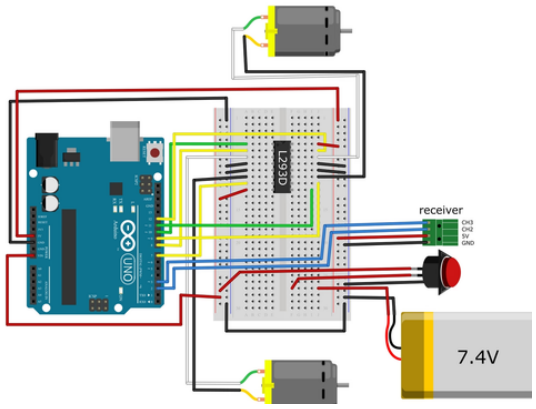
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MARINE DEBRIS RC BOAT







What?

- Built a **wireless aquatic robot** for debris removal and surface cleanup.
- Used **dual brushless motors** and **LiPo power** for propulsion.

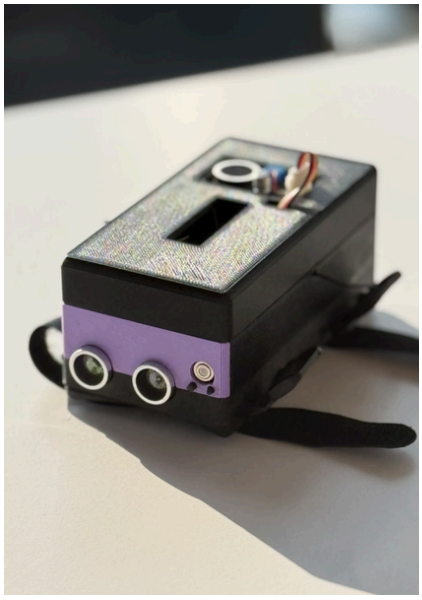
How?

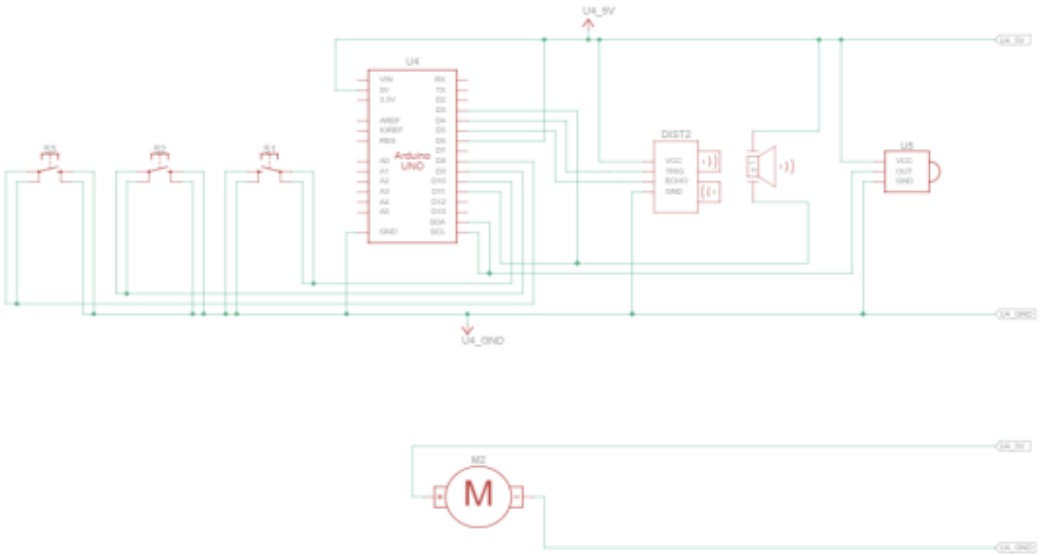
- Integrated **Arduino Uno** with **RF control** and **ESC-driven motors.**
- Designed waterproof housing and efficient power circuit for stability.

Results

- Achieved smooth wireless control and **strong thrust performance**.
- Ran **40+ minutes per charge** with reliable debris collection.

SEE-R OBJECT AND TEMPERATURE DETECTION DEVICE





What?

- Built **SEE-R**, a wearable device using **IR** and **ultrasonic sensors** for heat and obstacle detection.
- Powered by an **Arduino Uno** with vibration and audio feedback.
- Designed for lightweight, hands-free operation and user comfort.

How?

- **Prototyped circuits** and validated sensor integration via **TinkerCAD** and **breadboard testing**.
- Integrated all components with precise wiring, button extensions, and vibration module alignment.

Results

- Achieved **>90% detection accuracy** in real-world testing.
- Total weight under 300 g with stable sensor performance.
- Delivered a **fully functional prototype** with **reliable vibration** and **audio alerts**.

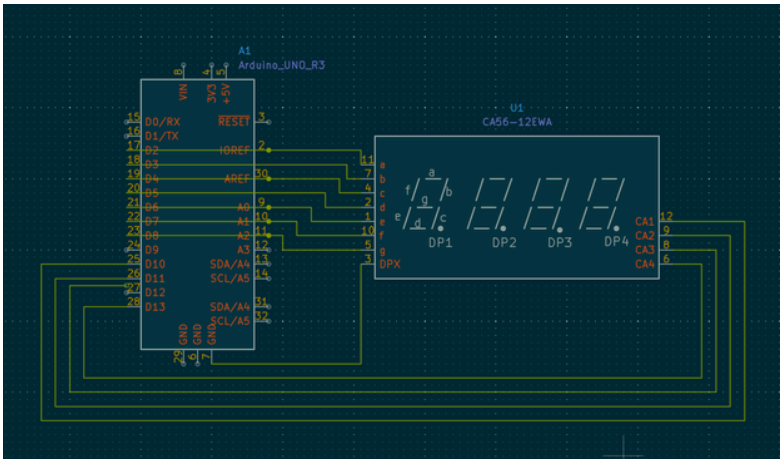
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HP3435A MULTIMETER REVERSE-ENGINEERING PROJECT





What?

- Reverse-engineered a vintage **HP3435A digital multimeter** with a **failed display system**.
- Built an **Arduino-based data acquisition system** to capture **voltage and current readings** in real time.

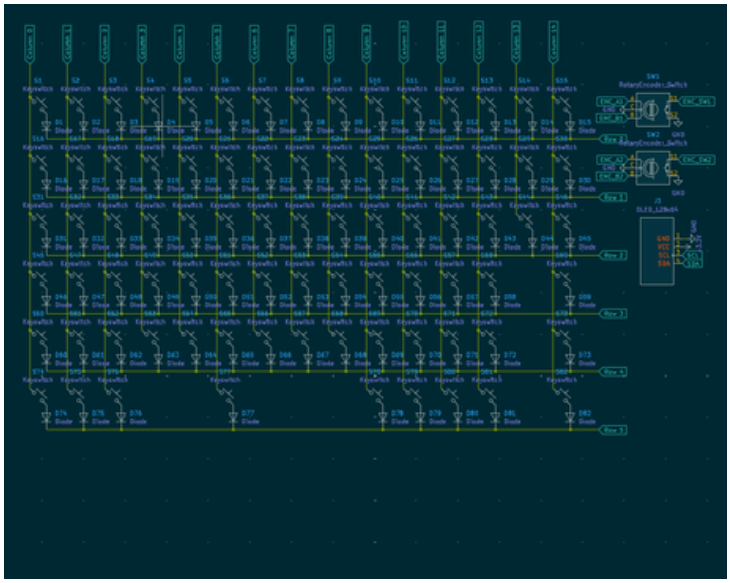
How?

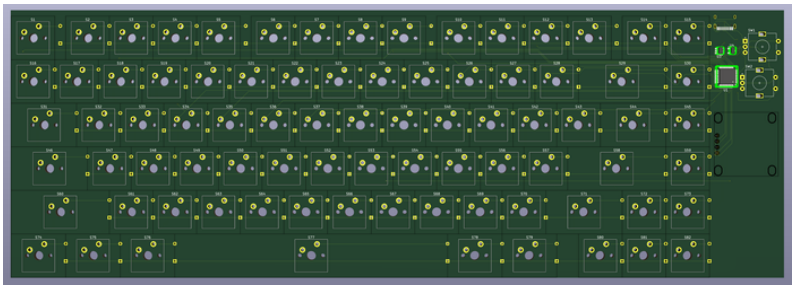
- **Interfaced Arduino Uno** with the **multimeter's display driver** via digital I/O mapping.
- Implemented **serial data logging to Excel** for real-time graphing and calibration comparison.

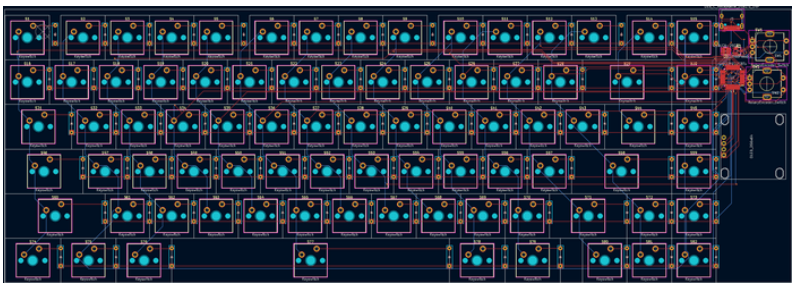
Results

- **Successfully restored measurement readouts** via serial monitor and Excel dashboard.
- **Validated readings** against reference equipment with **<2% deviation.**

STM32 BASED MECHANICAL KEYBOARD







What?

- Designed the PCB and **mechanical case** for a **custom** 82-key Hall-effect USB **keyboard**.
- Built around an **STM32 MCU**, with an **OLED display**, **dual rotary encoders**, and **USB-C**.

How?

- **Captured the schematic** and laid out the full **PCB in KiCAD.**
- **Designed the mechanical case**, aligning **mounting points** and **cutouts** for the USB-C port and OLED.

Results

- Completed a **fully routed PCB** covering all 82 keys plus the MCU, display, and encoder circuits.
- Delivered a **case design matched to the board outline**, so the PCB and enclosure fit together as one assembly..